

Correlated and Non-Correlated Subqueries in MySQL

Step 1: Create Sample Table

```
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    salary INT,  
    dept_id INT  
);
```

Step 2: Insert Sample Data

```
INSERT INTO employees (emp_id, name, salary, dept_id)  
VALUES  
(1, 'Ali', 5000, 1),  
(2, 'Sara', 7000, 1),  
(3, 'John', 6000, 2),  
(4, 'David', 8000, 2),  
(5, 'Emma', 4000, 3);
```

Table Data

emp_id	name	salary	dept_id
1	Ali	5000	1
2	Sara	7000	1
3	John	6000	2
4	David	8000	2
5	Emma	4000	3

1. Non-Correlated Subquery

Problem

Find employees earning more than the average salary.

Query

SELECT name, salary

FROM employees

WHERE salary >

(

SELECT AVG(salary)

FROM employees

);

Inner Query Execution

SELECT AVG(salary)

FROM employees;

Average salary:

6000

So main query becomes:

SELECT name, salary

FROM employees

WHERE salary > 6000;

Output

name	salary
Sara	7000
David	8000

Why Non-Correlated?

Because subquery:

```
SELECT AVG(salary) FROM employees
```

does NOT use outer query values.

It runs only once.

2. Correlated Subquery

Problem

Find employees earning more than the average salary of their own department.

Query

```
SELECT e1.name, e1.salary, e1.dept_id
FROM employees e1
WHERE e1.salary >
(
    SELECT AVG(e2.salary)
    FROM employees e2
    WHERE e2.dept_id = e1.dept_id
);
```

Understanding Step-by-Step

For Employee Ali

Outer row:

name	salary	dept_id
Ali	5000	1

Subquery becomes:

```
SELECT AVG(salary)
```

```
FROM employees
```

```
WHERE dept_id = 1;
```

Average = 6000

Check:

$5000 > 6000 \rightarrow \text{False}$

Ali not selected.

For Employee Sara

Subquery:

SELECT AVG(salary)

FROM employees

WHERE dept_id = 1;

Average = 6000

Check:

$7000 > 6000 \rightarrow \text{True}$

Sara selected.

For Employee David

Subquery:

SELECT AVG(salary)

FROM employees

WHERE dept_id = 2;

Average = 7000

Check:

$8000 > 7000 \rightarrow \text{True}$

David selected.

Final Output

name	salary	dept_id
Sara	7000	1

David	8000	2
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Main Difference

Feature	Non-Correlated	Correlated
Depends on outer query	No	Yes
Execution count	Once	Once per row
Performance	Faster	Slower
Uses outer query column	No	Yes

Easy Memory Trick

Non-Correlated

“Independent subquery”

SELECT AVG(salary) FROM employees

Runs alone.

Correlated

“Connected subquery”

WHERE e2.dept_id = e1.dept_id

Depends on outer query row.